SHAUNAK

Home work 10

1

X & Y are independent $\therefore$ $f_{XY}(x,y) = f_X(x)\, f_Y(y)$

$Y \sim \text{Uniform}(0,1)$

To show,

pdf of $Z = X + Y$ is

$f_Z(z) = F_X(z) - F_X(z-1)$

pdf $X = f_X(x)$

cdf $X = F_X(x)$.

cdf of Z $= F_Z(z) = P(Z \le z)$

$= P(X + Y \le z)$

Now $P(X+Y \le z) = \int_{-\infty}^{\infty} P(Y \le z - x \mid X = x)\, f_X(x)\, dx$ when we use TPL

Since X & Y are independent,

$$F_Z(z) = \int_{-\infty}^{\infty} P(Y \le z - x \mid X = x)\, f_X(x)\, dx$$

$$= \int_{-\infty}^{\infty} P(Y \le z - x)\, f_X(x)\, dx$$

$$= \int_{-\infty}^{\infty} F_Y(z-x)\, f_X(x)\, dx$$

$$\therefore \text{pdf } f_Z(z) = \frac{d}{dz} F_Z(z) = \int_{-\infty}^{\infty} \frac{d}{dz} F_Y(z-x)\, f_X(x)\, dx$$

$$= \int_{-\infty}^{\infty} f_X(x) f_Y(z-x)\, dx$$

$$f_Y(y) = \begin{cases} \frac{1}{b-a} & a \le y \le b \\ 0 & \end{cases}$$ here $a = 0$, $b = 1$

$$= \begin{cases} 1 & 0 \le y \le 1 \\ 0 & \end{cases}$$

so $0 \le z - x \le 1$ so $f_Y(z-x) = \begin{cases} 1 & 0 \le z - x \le 1 \\ 0 & \text{otherwise.} \end{cases}$

$\therefore z - 1 \le x \le z$

so $x = [z-1, z]$

$\therefore f_Z(z) = \int_{z-1}^{z} f_X(x) \cdot 1 \cdot dx$ since $f_Y(z-x) = 1$ within the range

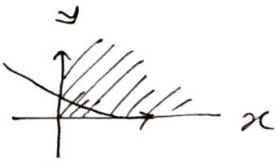
$f_Z(z) = F_X(z) - F_X(z-1)$

(2) joint density function of X and Y is given by,

$$f_{XY}(x,y) = \begin{cases} e^{-(x+y)} & x>0, y>0 \\ 0 & \text{otherwise.} \end{cases}$$

To find density func of rv $Z = X/Y$

Cdf of z, $F_Z(z) = P(Z \le z) = P\left(\frac{X}{Y} \le z\right)$



$x>0, y>0$, $x \le zy$

$y \in (0,\infty)$ where as $x \in [0, zy]$ as $0 \le x \le zy$

$$\therefore F_Z(z) = \int_{y=0}^{\infty} \int_{x=0}^{zy} e^{-(x+y)} dx\, dy \quad \text{—(1)}$$

inner integral $\int_{x=0}^{x=zy} e^{-(x+y)} dx = -\left[e^{-x}\right]_0^{zy}$

$= \left[-e^{-zy} + 1\right] e^{-y}$

$= e^{-y}\left[1 - e^{-zy}\right]$

now adding into outer integral (1)

$$F_Z(z) = \int_0^{\infty} e^{-y}\left[1 - e^{-zy}\right] dy$$

$$= \int_0^{\infty} e^{-y} dy - \int_0^{\infty} e^{-y} e^{-zy} dy \quad \text{—(2)}$$

now $\int_0^{\infty} e^{-y} dy = -\left[e^{-y}\right]_0^{\infty} = -0 + 1 = 1$

& $$\int_0^\infty e^{-(z+1)y}\,dy = \frac{1}{z+1}\left[e^{-z+1}\right]_0^\infty$$

$$= \frac{1}{z+1}\left[1-0\right]$$

$$= \frac{1}{z+1}$$

$\therefore$ (2) become,

$$F_Z(z) = 1 - \frac{1}{z+1} = \frac{z+1-1}{z+1} = \frac{z}{z+1}$$

now pdf $= \frac{d}{dz} F_Z(z) = \frac{d}{dz}\left(\frac{z}{z+1}\right)$

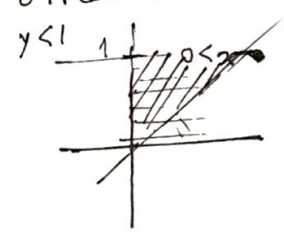
$$= \frac{(z+1)-z}{(z+1)^2} = \frac{1}{(z+1)^2}$$

$$\therefore \quad f_Z(z) = \begin{cases} 1/(z+1)^2 & z > 0 \\ 0 & \text{, otherwise} \end{cases}$$

Ans

(3) joint pdf of X & Y are

$$f_{XY}(x,y) = \begin{cases} c\,e^{-(x+y)} & 0 < x < y < 1 \\ 0 & \text{, otherwise} \end{cases}$$



where c is constant.

find pdf of $Z = X + Y$

$$F_Z(z) = P(Z \le z) = P(X + Y \le z)$$

$\therefore$ $0 < x < y < 1$, $x \in (0, y)$, $y \in (0,1)$

$$\int_0^1 \int_0^y f_{XY}(x,y)\,dx\,dy = 1 \quad \text{or} \quad \int_0^1 \int_0^y c\,e^{-(x+y)}\,dx\,dy = 1$$

(1)

inner integral

$$\int_0^y e^{-x}\,dx = 1 - e^{-y}$$

outer intergl with inner integral

$$c\int_0^1 e^{-y}(1-e^{-y})\,dy = c\int_0^1 [e^{-y} - e^{-2y}]\,dy$$

$$= c\left[-e^{-y}\right]_0^1 + c\frac{1}{2}\left[e^{-2y}\right]_0^1$$

$$= c\left[-e^{-1}+1\right] + \frac{c}{2}\left[e^{-2}-1\right]$$

$$= c\left[\frac{e^{-2}}{2} - e^{-1} + \frac{1}{2}\right]$$

$\therefore$ from condition ①

$$c = \frac{1}{\left[\frac{e^{-2}}{2} - e^{-1} + \frac{1}{2}\right]} = \frac{1}{\frac{1}{2}\left[e^{-2} - 2e^{-1} + 1\right]}$$

$$= \frac{2}{\left[e^{-1}-1\right]^2}$$

$$c = \frac{2}{\left[1-e^{-1}\right]^2}$$

Now to find $f_Z(z)$, $Z = X + Y$ has boundary

① $x < y \Rightarrow x < z - x \quad \therefore x < z/2$

② $y < 1 \Rightarrow z - x < 1 \quad \therefore x > z - 1$

& $x > 0$

$\therefore$ ~~when~~ $\max(0, z-1) < x < z/2$

Case 1 $\quad x \in [0, z/2]$

$$f_Z(z) = \int_0^{z/2} c e^{-z} dx$$

$$= c e^{-z} \left(x \right]_0^{z/2} = c e^{-z} [z/2] = \frac{c e^{-z} z}{2}$$

$$= \frac{e^{-z} \cdot z}{(1 - e^{-1})^2}$$

also $\quad x \in [z-1, z/2]$

$$f_Z(z) = \int_{z-1}^{z/2} c e^{-z} dx = c e^{-z} \left[\frac{z}{2} - (z-1) \right]$$

$$= c e^{-z} \left[\frac{z}{2} - z + 1 \right]$$

$$= c e^{-z} \left[\frac{z - 2z + 2}{2} \right]$$

$$= \frac{c e^{-z}}{2} \left[2 - z \right]$$

$\therefore$ finally substituting value of c & writing $f_Z(z)$ for all bounds $\quad 0 < z < 2$

$$f_Z(z) = \begin{cases} \dfrac{1}{(1-e^{-1})^2} \cdot e^{-z} & \text{when } 0 \leq z < 1 \\ \dfrac{1}{(1-e^{-1})^2} \cdot e^{-z}(2-z) & 1 < z < 2 \\ 0 & \text{otherwise.} \end{cases}$$

(4) Let X and Y two independent binomial r.v with pmf (5)

$$p_X(k) = p_Y(k) = \begin{cases} \binom{n}{k} p^k (1-p)^{n-k} & 0 \leq k \leq n \\ 0 & \text{otherwise} \end{cases}$$

Let $Z = X + Y$. To find pmf of Z.

$X \sim \text{Binomial}(n, p)$, $Y \sim \text{Binomial}(n, p)$

Sum $Z = X + Y$ is convolution

pmf of Z

Let $X = j$, $Y = k-j$

$$P(Z = k) = P(X + Y = Z)$$

$$= \sum_{j=0}^{k} P(X = j, Y = k-j)$$

Since X and Y are independent,

$$P(X = j, Y = k-j) = P(X = j) \cdot P(Y = k-j)$$

(1) becomes

$$\therefore P(Z = k) = \left[\sum_{j=0}^{k} \binom{n}{j} p^j (1-p)^{n-j}\right]\left[\binom{n}{k-j} p^{k-j} (1-p)^{n-(k-j)}\right]$$

$$= \sum_{j=0}^{k} \binom{n}{j}\binom{n}{k-j} p^{j+k-j} (1-p)^{n-j+n-k+j}$$

$$= \sum_{j=0}^{k} \binom{n}{j}\binom{n}{k-j} p^k (1-p)^{2n-k}$$

hence $\sum_{j=0}^{k} \binom{n}{j}\binom{n}{k-j} = \binom{2n}{k}$

this comes from Vandermonde's identity,

$$\sum_{j=0}^{k} \binom{r}{j}\binom{s}{k-j} = \binom{r+s}{k}$$

$\therefore \quad P(Z=z) = \binom{2n}{k} p^k (1-p)^{2n-k}$

Ans

where $0 \leq k \leq 2n$

$\therefore$ Result shows $P(Z=z)$ has Binomial distribution.

so when we sum two independent binomial random variables, it has same prob of success & the number of trial aggregate together $(n+n=2n)$ hence

so ~~Pr~~ $Z \sim \text{Binomial}(2n, p)$ Ans

5 $N \sim$ geometric (p)

$$p_N(k) = p(1-p)^k \quad , \quad k = 0, 1, 2, \ldots$$

$$0 \leq p \leq 1$$

$N \rightarrow$ total no of bytes in a message $\rightarrow$ r.v

$M \rightarrow$ fixed packet size not r.v

$Q \rightarrow$ no of full packets $\lfloor N/M \rfloor \rightarrow$ r.v

$R \rightarrow$ leftover $\rightarrow$ r.v.

N	M	Q $\lfloor N/4 \rfloor$	R
0	4	0	0
1	4	0	1
2	4	0	2
3	4	0	3

$$\therefore \check{N} = M\check{Q} + \check{R}$$

Joint pmf defined as $P(Q = q, R = r) = P(N = Mq + r)$

$$= p(1-p)^{Mq+r}$$

here $q \in \{0, 1, 2 \ldots\}$ and $r \in \{0, 1 \ldots M-1\}$

Marginal pmf of Q.

Sum over joint pmf over all possible values of R.

$$P(Q = q) = \sum_{r=0}^{M-1} P(Q = q, R = r)$$

$$= \sum_{r=0}^{M-1} p(1-p)^{Mq+r} = p(1-p)^{qM} \sum_{r=0}^{M-1} (1-p)^r$$

we know, $\sum_{i=0}^{n-1} x^i = \frac{1-x^n}{1-x}$, here $x = (1-p)$ & $n = r$

$$\therefore P(Q = q) = p(1-p)^{qM} \left(\frac{1-(1-p)^M}{1-(1-p)} \right)$$

$$= p(1-p)^{qM} \frac{1-(1-p)^M}{p}$$

$$= (1-p)^{qM} (1-(1-p)^M)$$

Marginal pmf of R

here we have to sum the joint pmf of all values of Q

$$\therefore P(R=r) = \sum_{q=0}^{\infty} P(Q=q, R=r)$$

$$= \sum_{q=0}^{\infty} p(1-p)^{Mq+r}$$

$$= \sum_{q=0}^{\infty} p(1-p)^{Mq}(1-p)^{r}$$

$$= p(1-p)^{r} \sum_{q=0}^{\infty} (1-p)^{Mq}$$

we know $\sum_{i=0}^{\infty} x^{i} = \frac{1}{1-x}$

here $x = (1-p)^{M}$ and $i = q$

$$\therefore P(R=r) = p(1-p)^{r} \frac{1}{1-(1-p)}$$

$$= (1-p)^{r} p \left(\frac{1}{1-(1-p)^{M}} \right)$$

$$= \cancel{(1-p)^{r}}$$

$$= p(1-p)^{r} \frac{1}{1-(1-p)^{M}}$$ Ans

6 $X(t) = e^{At}$ $\quad A \to$ crv

Mean f^n $\quad \mu_X(t) = E[X(t)] \; ; \forall t \in T$

$$= E[e^{At}]$$

$$= \int_{-\infty}^{\infty} x f_{X(t)}(x)\, dx \quad \text{—— (1)}$$

now we fix t first so $X = e^{At} = g(A)$

$x = g(a) = e^{at}$ $\quad \therefore$

or $\ln(x) = at \quad \therefore \quad a = \dfrac{\ln(x)}{t}$

$$\frac{da}{dx} = \frac{1}{t} \cdot \frac{1}{x}$$

$$\therefore f_{X(t)}(x) = f_A(a) \cdot \left|\frac{da}{dt}\right| = f_A\left(\frac{\ln(x)}{t}\right) \left|\frac{1}{xt}\right| \quad \text{—— (2)}$$

$\therefore$ (1) becomes

$$\mu_X(t) = \int_0^{\infty} \cancel{x} \cdot \left[f_A\left(\frac{\ln(x)}{t}\right) \frac{1}{\cancel{x}t} \right] dx \qquad \text{since } e^{At} > 0$$

$$= \int_0^{\infty} \frac{1}{t} f_A\left(\frac{\ln(x)}{t}\right) dx$$

to convert in terms of A, let $a = \dfrac{\ln(x)}{t}$ $\quad \therefore x = e^{at}$

$\dfrac{dx}{da} = t \cdot e^{at}$

as $x \to 0$, $a = \dfrac{\ln(x)}{t} \to -\infty$

$x \to \infty$, $a \to \infty$

$$\therefore \mu_{X(t)} = \int_{-\infty}^{\infty} \frac{1}{t} f_A(a) \left(t e^{at} da\right) = \int_{-\infty}^{\infty} e^{at} f_A(a)\, da$$

Ans

to estimate auto correlation function

$R_{XX}(t_1, t_2) = E[X(t_1) X(t_2)]$ by definition.

$X(t_1) X(t_2) = e^{At_1} \cdot e^{At_2} = e^{A(t_1+t_2)}$

$\therefore \quad R_{XX}(t_1, t_2) = E[e^{A(t_1+t_2)}]$

$$= \int_{-\infty}^{\infty} e^{a(t_1+t_2)} f_A(a)\, da \qquad \underline{\text{Ans}}$$

Next is first order density $f_{X(t)}(x)$ of rp $X(t)$

eqn (2) $\quad f_{X(t)}(x) = f_A\left(\frac{\ln(x)}{t}\right) \left|\frac{1}{xt}\right|, \quad x > 0$ and $t \neq 0$ $\underline{\text{Ans}}$

when $t = 0$, $\quad X(0) = e^{A \cdot 0} = 1$

$\therefore \quad f_{X(0)}(x) = \delta(x-1) \qquad \underline{\text{Ans}}$

(7) rv $C \sim \text{Uniform}(0, T)$ for $T \in \mathbb{R}$ (+ve constant).

$X(t) = u(t-C)$, $\quad u(t)$ is unit step fn.

to find auto corr fn.

$R_{XX}(t_1, t_2) = E[X(t_1) X(t_2)]$ by def^n —— (1)

$X(t_1) = u(t_1 - C)$

$X(t_2) = u(t_2 - C)$

$R_{XX}(t_1, t_2) = E[u(t_1 - C)\, u(t_2 - C)]$

$$= \cancel{E[u(t_1-c)]\,E[u(t_2-c)]}$$

So $u(t-c)$ is step f^n & takes values of 1 if $t-c \geq 0$

or $c \leq t$ & 0 if otherwise

$u(t_1-c) = 1$ if $c \leq t_1$

$u(t_1-c) = 1$ if $c \leq t_2$

so $u(t_1-c) \cdot u(t_2-c)$ to be equal both t_1 & t_2

must satisfy $c \leq t_1$ and $c \leq t_2$

Say $t_{min} = \min(t_1, t_2)$ $\therefore$ here product is 1 if $c \leq t_{min}$ & otherwise

~~what~~ $\cancel{R_{xx}(t_1,t_2) = E[x(t_1)x(t_2)]}$

$\therefore$

$\therefore$ $u(t_1-c)\,u(t_2-c)$

$= u(t_{min}-c)$

$\therefore$ $R_{xx}(t_1,t_2) = E[u(t_{min}-c)]$

Since $c \sim$ uniform $(0,T]$

$\therefore$ $R_{xx}(t_1,t_2) = P(c \leq \min(t_1,t_2))$

$\therefore$ cdf $R_{xx} = \begin{cases} 0 & \text{if } \min(t_1,t_2) \leq 0 \\ \frac{\min(t_1,t_2)}{T} & \text{if } 0 < \min(t_1,t_2) < T \\ 1 & \text{if } \min(t_1,t_2) \geq T \end{cases}$

$= 0$

Ans

8. $X(t) = A \sin \omega_0 t$ $\quad \omega_0 \to$ fixed freq.

$A \sim$ r.v. uniform $[-1, 1]$

$$f_A = \frac{1}{1+1} = \frac{1}{2}$$

(a) Mean f^n of $X(t) = E[X(t)]$

$$= E[A \sin \omega_0 t]$$

$$= E[A]\, E[\sin \omega_0 t] \qquad E[A] = \frac{a+b}{2} \cdot \cancel{E[\sin \omega_0 t]}$$

$$= \frac{-1+1}{2} = 0$$

$\therefore \quad \mu_{X(t)} = 0 \cdot \sin \omega_0 t$ | ω_0 & t are not r.v.

$= 0$ Ans

(b) Density function of r.v $X\left(\frac{\pi}{2\omega_0}\right)$

$$X\left(\frac{\pi}{2\omega_0}\right) = A \sin \omega_0 \cdot \frac{\pi}{2\omega_0} = A \sin\left(\frac{\pi}{2}\right)$$

$$= A \cdot 1 = A.$$

$$f_A = \frac{1}{2}$$

$$f_{X\left(\frac{\pi}{2\omega_0}\right)}(x) = \begin{cases} 1/2 & -1 \le x \le 1 \\ 0 & \text{otherwise} \end{cases}$$

Ans

9. (a) If heads (H), $X(t) = \sin 2\pi t$

if tails (T), $X(t) = 2t$

$P(H) = \frac{1}{2}$ & $P(T) = \frac{1}{2}$

if faire die,

$\therefore \quad X(t|H) = \sin 2\pi t$

& $X(t|T) = 2t$

$E[X(t)] = \sum X(t|H)$.

$$E[X(t)] = \sum_x x \, P(t \mid \text{coinflip state})$$

$$= \frac{1}{2} \sin(2\pi t) + \frac{1}{2}(2t)$$

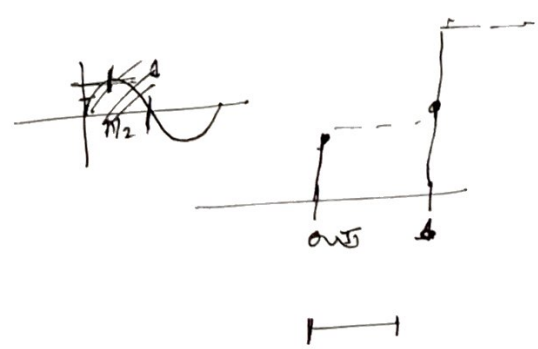
$$= \frac{1}{2} \sin(2\pi t) + 2t$$

$$= t + \frac{1}{2} \sin(2\pi t)$$

a. (b) cdf of $F_{X(t)}(x)$ for $t = 0.25$, $t = 0.5$ & $t = 1$

Generally $X(t)$ taking values

$$F_{X(t)}(x) = P(x(t) \le x).$$



when $t = 0.25$

if H: $X = \sin\left(2\pi \cdot \frac{1}{4}\right) = \sin\left(\frac{\pi}{2}\right) = 1.0$

if T: $X = 2t = 2 \cdot \frac{1}{4} = \frac{1}{2} = 0.5$.

$$\therefore F_{X(0.25)}(x) = \begin{cases} 0 & x < 0.5 \\ 1/2 & 0.5 \le x < 1 \\ 1 & x \geq 1 \end{cases}$$

$t = 0.5$

if H: $X(0.5) = \sin\left(2\pi \cdot \frac{1}{2}\right) = \sin \pi = 0$

if T: $X(0.5) = 2t = 2 \cdot \frac{1}{2} = 1$

$$F_{X(0.5)}(x) = \begin{cases} 0 & x < 0 \\ 1/2 & 0 \le x < 1 \\ 1 & x \geq 1 \end{cases}$$

$F_{X(1)}(x)$

if $H : X(1) = \sin(2\pi) = 0$

$T \quad X(1) = 2$

$$\therefore F_{X(1)}(x) = \begin{cases} 0 & x < 0 \\ 1/2 & 0 \leq x < 2 \\ 1 & x \geq 2 \end{cases}$$

Ans

10 $X(t)$ is WSS & zero-mean & gaussian process

(a) auto corr $R_X(\tau) = 4e^{-2|\tau|}$

A r.p is WSS if

$\mu_X(t) = \text{Const} \quad \forall t$, here is zero.

& $R_X(t_1, t_2) = R_X(t_1 - t_2) \quad \forall t_1, t_2$

$= 4e^{-2|\tau|}$ so WSS $\tau = |t_1 - t_2|$

at $t = 3$

$X(3) \sim N(0, \sigma_X^2)$

$\mu_X = E[X(t)] = 0$ given.

~~$R_X(t_1, t_2) = E[X(t_1) X(t_2)]$~~

~~$= E[X(3) X(3)]$~~

~~$= E[(X(3))^2]$~~

~~$R_X(3) = 0$~~

~~$E[X(3)] = 0$~~

~~$Var[X(3)] = 0$~~

$R_X(\tau) = E[X(t) X(t+\tau)]$

$\tau = 0$

$= E[X^2(t)]$

$R_X(0) = E[X^2(t)]$

$\therefore \sigma_X^2 = 4e^{-2|0|} = 4 \quad \therefore \sigma_X = 2$

$\therefore Var(X)$

$= E[X^2] - (E[X])^2$

$= R_X(0) - (\mu_X)^2$

$\sigma_X^2 = R_X(0)$

$\therefore \quad X(t) \sim N(0,4)$ distribution

to calculate prob.,

$P(X(t) \leq 3)$, we standardize into std normal variable $Z \sim N(0,1)$ using $Z = \frac{X - \mu_X}{\sigma_X}$

$$\therefore P(X(t) \leq 3) = P\left(\frac{X(t) - 0}{2} \leq \frac{3-0}{2}\right)$$

$$= P(Z \leq 1.5)$$

$$\therefore P(X(t) \leq 3) = \Phi(1.5)$$

$$= 0.93 \quad \underline{\text{Ans}}$$

(b) $E[(X(t+1) - X(t-1))^2]$

$$E[X^2(t+1) + X^2(t-1) - 2X(t+1).X(t-1)]$$

$$= E[X^2(t+1)] + E[X^2(t-1)] - 2E[X(t+1)].[X(t-1)]$$

$$= E[X^2(t+1)] + E[X^2(t-1)] - 2E[X(t+1).X(t-1)] \quad \text{—— (1)}$$

by def^n $R_X(\tau) = E[X(t)X(t+\tau)]$

first term $E[X^2(t+1)] = R_X(0)$ $\qquad \tau = (t+1) - (t+1) = 0$

second term $E[X^2(t-1)] = R_X(0)$ $\qquad \tau = (t-1) - (t-1)$

third term

$E[X(t+1).X(t-1)] = R_X(2)$ $\qquad \tau = (t+1) - (t-1) = 2$

RHS of eq^n (1) becomes

$$= R_X(0) + R_X(0) - 2.R_X(2)$$

$$= 4 + 4 - 2.(4e^{-2|2|}) = 8 - 8e^{-4}$$

$$= 8(1 - e^{-4}) \quad \underline{\text{Ans}}$$

(11) $X(t)$ is wss. zero mean. gap

to show if (17)

$Z(t) = X^2(t)$

$\therefore Z(t)$ is wss

& $C_Z(\tau) = 2C_X^2(\tau)$

① $E[X(t)] = \mu_X$

② $R_{XX}(t_1, t_2) = R_X(\tau)$ $\quad \tau = t_2 - t_1$

also $R_X(\tau) = E[X(t)\, X(t+\tau)]$ —— ①

$\therefore E[Z(t)] = E[X^2(t)]$

from ① $R_X(\tau=0) = E[X^2(t)] = R_X(0)$

$\therefore E[Z(t)] = E[X^2(t)] = R_X(0) = \text{Constant}$

$= C_X(0) = R_X(0)$

and do not depend on t.

② $R_{ZZ}(t_1, t_2) = E[Z(t_1)\, Z(t_2)]$

$= E[X^2(t_1)\, X^2(t_2)] = E[X^2(t)\, X^2(t+\tau)]$

apply this, $E[X_1^2 X_2^2] = E[X_1^2]\, E[X_2^2] + 2\left(E[X_1 X_2]\right)^2$

$E[X^2(t)\, X^2(t+\tau)] = R_X(0) \cdot R_X(0) + 2 \cdot \left(R_X(\tau)\right)^2$

$\therefore$

$= R_X(0)\, R_X(0) + 2\left[R_X(\tau)\right]^2$

$= C_X(0) \cdot C_X(0) + 2\, C_X^2(\tau)$

$R_Z(\tau) = C_X(0)^2 + 2\, C_X^2(\tau)$ depends only on τ

$C_Z(\tau) = R_Z(\tau) - \left(E[Z(t)]\right)^2$

$= \left[C_X(0)^2 + 2C_X(\tau)\right] - C_X(0)^2$

$\therefore C_Z(\tau) = 2\, C_X^2(\tau)$

$\therefore$ ① mean $Z(t)$ is constant

② autocovariance & autocorrelation depends only on τ so wss.